

Validated Read-Only Geometric Stress Testing of Native Assignment States in Aerial Tiny-Object Detection

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ABSTRACT

Many NMS-free dense detectors retain one loss-active candidate for one-to-one (O2O) supervision, but the local geometry of this discrete assignment state is poorly characterised. We present a read-only geometric-stress audit that replays the native assigner while freezing model outputs, parameters, candidate grids, object size, and non-focal ground truths. Its implementation was qualified by analytical oracles, property invariants, mechanism mutations, three-way native/production/NumPy parity, and an independent continuous-boundary reference, with zero observed mismatch in the respective validation sets. On AI-TOD-v2, fixed one-pixel replay produced an O2O 8–16 minus 16–32 px scale contrast of 14.37 percentage points (pp), whereas equivalent-side replay reduced it to 0.13 pp and exposed an O2M contrast of -17.45 pp. Locking base eligibility removed 79.44% of the fixed O2O gap. First-divergence anatomy shifted from 67.09% eligibility boundaries under fixed stress to 51.01% under normalized stress, while within-set geometric reversals increased from 30.13% to 45.37%. Continuous analysis yielded restricted mean stability radii of 0.11992 for O2O and 0.10055 for assigned-positive O2M states. Native margin was more strongly associated with observed fixed-pair rank-boundary radius ($\rho = 0.628$) than with eligibility-boundary radius ($\rho = 0.153$). The results characterise assignment instability as a stress-contract-dependent property of native assignment-state boundary geometry, not a universal scale law, calibrated failure probability, or replacement assigner.

1. Introduction

Aerial tiny-object detection places discrete training decisions under unusually coarse geometric resolution. Dense candidates lie on stride-dependent lattices, while many NMS-free detectors use a one-to-one (O2O) branch that retains a single loss-active candidate for each ground-truth object [6, 7, 9, 10]. A small centre displacement can therefore change not only overlap or score, but also the identity, rank, eligibility, or positive-set membership used to construct supervision. These are properties of the native assignment state rather than postprocessed detections.

Existing tiny-object methods modify localisation similarity, receptive-field-aware assignment, scale-aware labels, feature resolution, or supervision density [1, 2, 3, 4, 15, 16, 17]. Dense-detector assignment methods likewise change candidate selection through adaptive matching, optimal transport, task alignment, or differentiable assignment [18, 19, 20, 21, 22, 23]. Such methods address how positives should be constructed. They do not, by themselves, measure how stable the resolved native assignment state is after training, nor whether an apparent scale response depends on the geometric stress used to probe it.

We study that measurement problem directly. The focal ground-truth centre is replayed under a fixed-pixel or object-normalized displacement while image pixels, extracted features, model parameters, candidate grids, box size, and all non-focal ground truths remain fixed. This design isolates the response of the frozen native assigner. It also makes

the measurement contract explicit: a one-pixel shift asks about sensitivity to an absolute input displacement, whereas equivalent-side replay asks about the same fraction of object extent. Neither contract is treated as a correction for the other.

Three challenges follow. First, the instrument must reproduce native eligibility, Top- k selection, conflict resolution, tie handling, and numerical precision closely enough that a reported identity change is not a tracing artefact. Second, O2O identity, O2M rank, and O2M assigned-positive-set membership are different states; collapsing them into one candidate-change label obscures what moved. Third, five-state replay observes a response at one displacement, but does not show how close the first boundary lies, whether a trajectory is right-censored, or which native stage produced the first divergence.

We address these challenges with a qualified read-only measurement implementation. Analytical oracles, property invariants, mechanism mutations, native/production/NumPy parity, and an independent continuous-boundary reference test the implementation before scientific interpretation. The audit then combines fixed and normalized replay, eligibility locking, first-divergence anatomy, O2O pre-resolution controls, O2M rank-set decomposition, and continuous stability-radius analysis. Weighted Kaplan–Meier and Aalen–Johansen summaries retain censoring and competing boundary causes. Margin is evaluated as a local rank-separation coordinate, while spillover, reduced-budget screening, and predictive-shape analyses define the diagnostic boundary.

The contributions are fourfold:

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- **Qualified native-state measurement.** We provide a loss-faithful, read-only audit whose native-path implementation is checked by analytical, mutation, differential-parity, and independent continuous-boundary references.
- **Stress-contract and pathway structure.** Fixed-pixel O2O scale responses are largely coupled to eligibility boundaries; normalized stress reduces that component and increases the share of within-set geometric rank reversals, while O2O–O2M branch divergence remains present across the audited magnitudes, datasets, archived stages, and a compatible second detector contract.
- **Assignment-state boundary geometry.** Continuous boundary analysis, O2O pre-resolution control, and O2M rank-set decomposition distinguish loss-active identity stability, pre-conflict rank stability, and assigned-positive-set persistence.
- **Construct and diagnostic boundaries.** Native margin localises fixed-pair rank separation more strongly than eligibility distance, whereas nonlinear margin shape, spillover, and reduced-budget analyses delimit what the audit can support. The framework is a measurement tool, not a replacement assigner, runtime uncertainty estimator, or validated treatment.

2. Related work

2.1. Tiny-object geometry and scale-aware assignment

NWD, RFLA, SimD, dynamic coarse-to-fine learning, and normalized Gaussian label assignment reduce the brittleness of overlap-based matching for tiny objects [1, 2, 3, 4, 15]. Recent scale-aware assigners further use virtual-anchor calibration or normalized Bhattacharyya distance [16, 17]. UGS addresses small-object localisation gradients rather than assignment identity [5]. DALA instead separates dense and sparse objects using ground-truth spatial density [13]. Sparse objects receive decreasing positive supervision during training, whereas dense objects retain O2O assignment. These methods modify geometry, optimisation, positive construction, or supervision density. Our study freezes the native assigner and audits the active identity produced by it.

More broadly, dense-detector assignment now includes adaptive and probabilistic candidate selection [18, 19], optimal transport and task alignment [20, 21], and differentiable matching [22, 23]. These methods optimise which candidates become positives; they do not measure the post-training stability of one native O2O loss-active identity.

2.2. End-to-end dense detection and dual supervision

DeFCN introduced prediction-aware O2O assignment with an O2M auxiliary objective [6]. OneNet used class-aware matching to enlarge the separation required for non-redundant prediction [7]. EOOD combined an oriented O2O assigner, gap regularisation, and O2M auxiliary heads [24]. DATE restored O2M supervision to accelerate O2O convergence [8]. Hybrid methods add grouped or auxiliary

supervision during training while retaining one-to-one inference [25, 26, 27]. PaQ-DETR couples dynamic queries with quality-aware O2M assignment [28]. YOLOv10 aligned the O2M and O2O criteria [9], whereas YOLO26 combined dual heads with NMS-free inference, direct LTRB regression, Progressive Loss, and small-target-aware positive coverage [10]. These methods improve assignment or supervision construction. We instead audit a native loss-active identity after training.

2.3. Different meanings of assignment ambiguity

Stable-DINO studied inter-layer Hungarian-matching instability, and many-to-one pedestrian matching challenged arbitrary selection among similar proposals [29, 30]. Both changed training to improve robustness. TCD linked assignment ambiguity to inconsistent preset intermediates and model predictions [11]. Its dynamic sample set addresses conventional single-stage assignment. DS-Det located query ambiguity in shared-weight decoders trained with O2M and O2O objectives [12]. O2F assigned dynamic soft supervision to several ambiguous anchors [31]. Output-level diagnostic tools such as TIDE decompose detection errors without accessing the training assignment path [32]. Our estimand retains one native O2O winner and measures its local stability. Existing assignment studies mainly replace matching or supervision rules. Our contribution is a post-training measurement of the native loss-active identity, not a replacement assigner.

2.4. Post-training neural-system testing and diagnostics

DNN testing studies how to select informative test inputs, estimate fault-revealing behaviour, and reduce labelling cost [35, 36]. Other diagnostic work uses near-miss concepts for model comparison and debugging [37], or treats anomalous behaviour as dependent on the evolving training state [38]. These studies analyse prediction behaviour, representations, test selection, or training anomalies. Our audit instead targets the resolved native assignment state used to construct dense-detector supervision and measures its response while network outputs remain frozen.

3. Methods

3.1. Study design and frozen detector contracts

The study is a post-training diagnostic analysis of native assignment state. The principal contract was a YOLO26s detector with native O2M and O2O task-aligned assignment [10]. A compatible YOLOv10-S contract provided a second-detector replication [9]. AI-TOD-v2 was the principal dataset and VisDrone was used for cross-dataset sensitivity [33, 34]. All replay analyses used frozen checkpoints. No replay result was used to update weights, alter inference, or select a replacement checkpoint.

The discovery sample comprised 300 outcome-blind stratified AI-TOD-v2 validation images. Dataset-specific sampling weights were the inverse image-inclusion probabilities. Unless stated otherwise, uncertainty was estimated

with 5,000 bootstrap replicates that resampled image clusters within the frozen sampling strata. The statistical unit is stated for every estimand because object-level fragility, directional transitions, candidate ranks, positive sets, and image-pair spillover are not interchangeable.

3.2. Native O2O identity, O2M states, and margin

For ground-truth object g , candidate i , and class c_g , native task alignment used

$$q_{ig} = p_{i,c_g}^{0.5} [\max\{\text{CIoU}(b_i, g), 0\}]^6, \quad (1)$$

Equation (1) fixes the candidate score used by every replay state. where p_{i,c_g} is the frozen sigmoid class score and b_i is the frozen decoded box. Candidate eligibility followed the detector's native small-target-aware region. The O2O path retained its native pre-conflict Top-7 set, resolved multi-GT candidate claims, and then produced one loss-active identity. If the post-resolution foreground and target-index state contained exactly one candidate for g , its index was denoted i_g^{active} .

When an active identity and a positive alternative were both available, the native runner-up $i_g^{(2)}$ was the highest- q positive alternative for the same object. The active-relative margin was

$$m_g = \frac{q_{i_g^{\text{active}},g} - q_{i_g^{(2)},g}}{q_{i_g^{\text{active}},g} + 10^{-12}}. \quad (2)$$

Equation (2) was evaluated only when both identities existed. Margin was undefined when the active identity or a positive alternative was absent. Undefined competition was reported as a separate structural state and was never imputed as an extreme margin.

O2M was represented by several distinct states. The legacy eligible Top-1 identity is the largest- q eligible candidate before native positive selection. The native pre-conflict identity is the largest- q candidate inside the native Top-10 set. The final assigned-positive set $\mathcal{P}_g$ contains candidates assigned to g after conflict resolution, and the assigned-set Top-1 identity is the largest- q element of $\mathcal{P}_g$. Rank state and set state were analysed separately.

3.3. Geometric stress and five-state replay

Only the focal ground-truth centre was moved. Image pixels, extracted features, parameters, candidate grids, decoded candidate boxes, focal width and height, and all non-focal ground truths remained fixed. For a base centre (x_g, y_g) , fixed one-pixel replay used

$$\mathcal{D}_{\text{px}} = \{(0, 0), (\pm 1, 0), (0, \pm 1)\}. \quad (3)$$

Equivalent-side replay used $\kappa = 0.0625$ and

$$\delta_g = \kappa \sqrt{w_g h_g}, \quad \mathcal{D}_{\text{eq}} = \{(0, 0), (\pm \delta_g, 0), (0, \pm \delta_g)\}. \quad (4)$$

Equations (3) and (4) define the two stress contracts. Sensitivity analyses used $\kappa \in \{0.03125, 0.0625, 0.125\}$. Fixed

and normalized contracts answer different perturbation questions and were not pooled.

For a native state $z_g(d)$ under displacement d , an object was stable when all legal shifted states equalled the base state and fragile when at least one legal shifted state differed. An unavailable base identity, missing active identity, or absence of a valid relative competitor was retained as undefined according to the relevant estimand. Directional analyses used each legal base-to-shift transition as the unit.

The downstream diagnostic previously termed FN@IoU50 is a coverage-based miss: it indicates that no frozen same-class prediction among the prespecified confidence and top-count outputs overlaps the focal GT by at least 0.5. Predictions are not consumed one-to-one, so this outcome is not the standard evaluator false negative.

Figure 1 summarises the resulting measurement object: a native loss-active identity is read from the unmodified training path, the focal GT alone is replayed through controlled centre shifts, and identity, eligibility, pathway, and continuous-boundary readouts are recorded without changing inference or supervision.

3.4. Instrument qualification and numerical contract

Instrument qualification preceded scientific interpretation. Six analytical oracle cases represented stable assignment, eligibility entry and exit, Top- k transition, conflict reassignment, rank reversal, and disappearance. Seven property invariants checked translation, GT-order, and state-consistency behaviour. Six prespecified mutations tested whether the validation suite detected incorrect eligibility, ranking, conflict, tie, and state logic.

Native execution, the production tracer, and an independent NumPy reference were compared stage by stage. The broader deterministic parity set contained 300 base states and 5,251 directional states. Continuous-boundary validation used a deterministic 30-image subset, one network forward per image, and an independent NumPy scanner that did not import production taxonomy helpers.

Native float32 execution was normative. Candidate ordering, exact ties, eligibility comparisons, clipped-CIoU zero handling, margin epsilon, and boundary brackets followed the frozen numerical contract. Numerical differences inside a prespecified tolerance were labelled boundary cases rather than silently assigned to a scientific pathway.

3.5. Eligibility and first-divergence analyses

The eligibility-lock counterfactual held the base eligibility state fixed during replay and recomputed the remaining native path. It estimates how much of the observed response depends on crossing hard eligibility boundaries under that counterfactual; it is not a training intervention. The complementary eligibility-stable analysis conditioned on objects or directions whose relevant eligibility state remained comparable in both replay states. Because this conditioning is observed after replay, it changes the estimand and may induce selection.

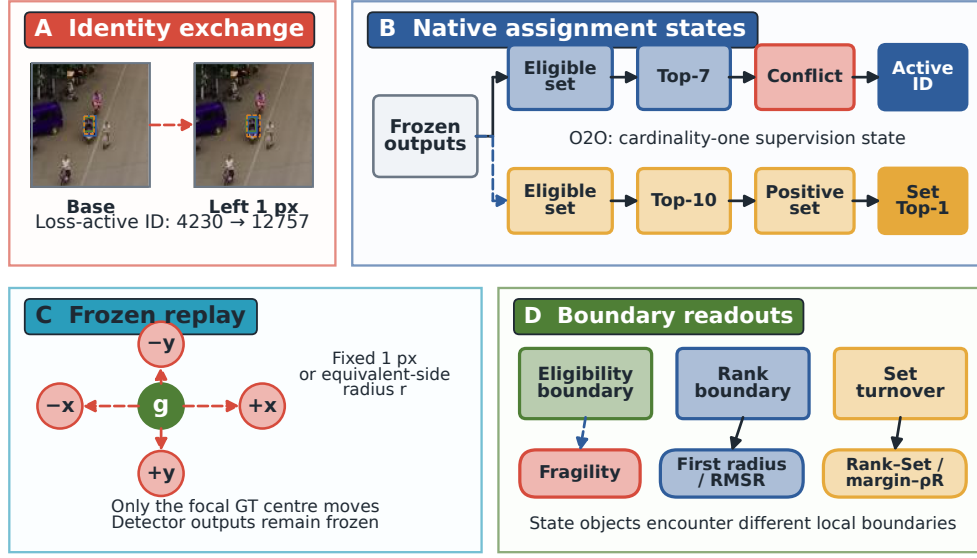


Figure 1: Read-only measurement of native assignment-state boundary geometry. (A) A frozen qualitative example in which the loss-active identity changes after a one-pixel focal-GT shift. (B) The audited O2O loss-active and O2M rank-set states remain native to their respective assignment paths. (C) Five-state replay moves only the focal GT centre while image pixels, features, parameters, candidate grids, decoded boxes, object extent, and all non-focal GTs remain fixed. (D) The audit returns discrete identity stability, first-divergence pathway, and continuous boundary-radius readouts; undefined competition remains a separate state. The example is explanatory and is not used as a detector output claim.

Every directional O2O identity exchange received one mutually exclusive first-divergence label under the instrumented audit hierarchy: eligibility boundary, Top-7 membership transition, conflict reassignment, within-set geometry rank reversal, active disappearance, or compound/other. The label is the earliest recorded native state difference, not a causal attribution. Clipped-CIoU-to-zero events were retained separately as overlapping geometry-clipping diagnostics.

For the pre-resolution control, the native pre-conflict Top-7 q winner was compared with the final post-conflict loss-active identity under fixed and equivalent-side replay. Scale-gap and object-fragility disagreements were computed within the O2O branch. Disagreements were then localised to the instrumented conflict stage.

3.6. O2M rank-set decomposition

For the base and shifted assigned-positive sets, exact equality, Jaccard similarity, and retention of the base set were recorded. Jaccard was defined only when the union was non-empty; empty/empty remained undefined rather than being assigned one. Base retention was undefined when the base set was empty. Empty-to-nonempty and nonempty-to-empty transitions were reported separately.

Assigned-set Top-1 changes were decomposed into rank-only exchange, set turnover with rank exchange, and set turnover without rank exchange. No post hoc Jaccard threshold was used. Fixed and equivalent-side contracts used the same image clusters, inverse inclusion weights, and 5,000 stratified image-cluster bootstrap replicates.

3.7. Continuous assignment-boundary geometry

Equivalent-side radius r was scanned from zero to $r_{\max} = 0.25$ in four cardinal directions. A forward coarse scan with step $1/64$ located the first observed same-to-changed bracket, and a forward fine scan with step $1/512$ refined that bracket. The algorithm did not assume monotonicity and therefore retained return transitions such as $A \rightarrow B \rightarrow A$. No GT clipping was allowed; trajectories whose base box was outside the legal domain were excluded under the frozen contract.

For state type k , the first observed boundary radius was ρ_{gdk} . Trajectories with no observed boundary through $r_{\max}$ were right-censored. Eligibility, Top- k , conflict, within-set rank, and other causes were treated as competing first boundaries where applicable. Weighted Kaplan-Meier survival $S_k(r)$ summarised overall state persistence, and Aalen-Johansen cumulative incidence summarised boundary causes.

The restricted mean stability radius was

$$\text{RMSR}_k(\tau) = \int_0^\tau S_k(r) dr, \quad \tau = 0.125. \quad (5)$$

Equation (5) was evaluated without a proportional-hazards assumption.

For fixed base active candidate A and runner-up B , ρ_R was the first observed radius at which their q order crossed while both remained comparable. An earlier eligibility, pre-Top- k , or conflict change was a competing censor; no crossing by $r_{\max}$ was a right censor. Margin- ρ_R association was reported only among observed crossings and was compared with the association between margin and observed eligibility-boundary radius. It is not a censoring-adjusted population correlation.

3.8. Spillover and secondary diagnostic analyses

Assignment spillover measured whether moving focal GT g changed the loss-active identity of a non-focal GT h . The primary structural factor was whether g and h shared at least one native pre-conflict candidate claim in the base or shifted state. Pair-level rates used inverse image-inclusion weights and 5,000 stratified image-cluster bootstrap replicates. The full 300-image analysis was run only after a prespecified 30-image pilot met the gated scientific-value criterion.

Margin-shape sensitivity compared base covariates plus a linear margin term with the same covariates plus a restricted cubic spline margin basis. Five image-grouped out-of-fold splits were used. Spline knots were estimated only from each training fold and then applied unchanged to held-out image groups. ROC-AUC, PR-AUC, and log loss were reported for identity fragility and coverage-based miss. This analysis tests shape sensitivity, not whether margin is a causal or optimal predictor.

Reduced-budget analysis drew 1,000 outcome-blind stratified image subsamples at budgets 25, 50, 100, 150, and 200. It reported absolute error relative to the full normalized paired contrast, sign recovery, and branch-order recovery. No nested bootstrap was run inside each subsample. A separate pre-access planning analysis reported discovery-bootstrap standard deviations and expected interval half-widths; it was not a power calculation.

3.9. Evidence governance and the attempted prospective audit

After discovery and instrument qualification, the replay implementation, numerical contract, sampling, taxonomy, bootstrap procedures, checkpoint identity, and analysis code were committed in an isolated repository and tagged before lockbox selection. A disjoint 300-image outcome-blind AI-TOD-v2 sample was then selected with frozen stratum allocation and zero overlap with the discovery images. Four raw extraction roles were completed under sealed mode.

The preregistered Tier-A family ordered three one-sided hypotheses by hierarchical gatekeeping: normalized paired branch divergence, normalized O2M scale contrast, and

fixed-minus-normalized O2O scale contrast. Tier B contained fixed-minus-normalized eligibility share and normalized-minus-fixed within-set share with Holm correction. The single permitted analyzer invocation wrote an irreversible outcome-access receipt before reading raw outcomes.

That invocation terminated before producing any H1–H5 estimate or verdict. The frozen analyzer required every selected image to appear in each raw table before its specified zero-cluster merge, although one selected image had no raw GT row. It also encountered three fixed-only common-valid GT keys. In accordance with the one-time rule, the prospective analysis was not rerun; no normative lockbox verdict exists. A corrected post-access analysis retained all 300 image clusters and used the 7,243-key fixed/normalized intersection. Those results are reported only as exploratory in the Supplementary Material and do not confer prospective status on any claim.

OpenAI Codex assisted code drafting, statistical-workflow implementation, manuscript editing, and figure-layout development. All generated code and outputs were reviewed by the authors, executed against frozen inputs, checked through deterministic reruns and independent validators, and reconciled with content-addressed manifests before manuscript use.

4. Results

4.1. Qualification of the native replay audit

The analytical suite recovered all six oracle cases and all seven property invariants. All six prespecified mutations were detected. Native execution, production tracing, and the independent NumPy reference produced zero mismatch across 300 base states and 5,251 deterministic directional states. In the separate continuous-boundary qualification, the independent NumPy implementation reproduced all 4,408 production trajectories in the 30-image subset, with zero event-radius difference and one network forward per image. These results qualify the measurement implementation; they are not detector-effect estimates.

4.2. Stress parameterisation exposes branch-specific response structure

On the AI-TOD-v2 seed-0 discovery sample, fixed one-pixel replay produced an O2O 8–16 minus 16–32 px fragility contrast of 14.37 pp. Under equivalent-side replay, the O2O contrast was 0.13 pp and its interval included zero. O2M behaved differently: its fixed contrast was 1.42 pp and uncertain, whereas its normalized contrast was -17.45 pp. Consequently, the paired O2O-minus-O2M contrast was positive under both contracts (Table 1). Fixed and normalized estimates were not pooled because the displacements answer different questions.

Across the three-point equivalent-side grid, all six O2M contrasts were negative and all six paired branch contrasts were positive. On AI-TOD-v2, paired contrasts ranged from 17.27 to 18.19 pp; on VisDrone they ranged from 34.79 to 37.25 pp. Absolute O2O contrasts varied with both κ and dataset: the AI-TOD-v2 intervals at $\kappa = 0.03125$ and

Table 1

Primary AI-TOD-v2 stress-contract contrasts in percentage points. Brackets are 95% stratified image-cluster bootstrap intervals. Positive values denote higher 8–16 px than 16–32 px fragility. These are discovery estimates.

Stress contract	O2O scale gap	O2M scale gap	Paired gap
Fixed 1 px	14.37 [10.84, 17.78]	1.42 [-3.18, 6.31]	12.95 [6.52, 19.16]
Equivalent-side $\kappa = 0.0625$	0.13 [-3.41, 3.15]	-17.45 [-22.72, -11.71]	17.58 [11.06, 23.98]

0.0625 included zero, whereas the VisDrone contrasts were positive at all three magnitudes. The paired direction was therefore stable over the tested grid, but the absolute O2O scale response was not a universal scale law.

The compatible YOLOv10-S contract showed the same normalized branch pattern: O2O was 0.94 pp with an interval spanning zero, O2M was -23.50 pp, and the paired contrast was 24.44 pp. This is replication across two compatible detector contracts, not architecture invariance.

The branch pattern was present at all six physically archived stages spanning the first completed epoch to the final checkpoint. Across those stages, mAP_{50:95} changed from 0.112 to 0.280, while fixed paired contrasts ranged from 11.84 to 15.09 pp and normalized paired contrasts from 17.34 to 21.35 pp. No checkpoint was available between epochs 100 and 280, so no continuity or interpolation is claimed across that interval.

Figure 2 brings these stress-contract, magnitude, archived-stage, and detector-contract results together. The shaded interval in panel C marks missing archived checkpoints and is not an interpolated trajectory.

4.3. Eligibility boundaries account for most of the fixed-pixel O2O scale gradient

Holding base eligibility fixed reduced the fixed O2O scale gap by 13.58 pp, corresponding to 79.44% [70.51%, 91.92%] of the native gap. The normalized reduction was 3.15 pp [1.42, 4.83]. Thus, the apparent fixed-pixel O2O gradient was strongly coupled to hard eligibility boundaries, although this structural counterfactual does not establish a complete causal mechanism.

The observational eligibility-stable subset retained 6,217 of 7,434 fixed-replay objects and 6,999 of 7,432 normalized-replay objects. Its fixed O2O scale contrast was 3.36 pp [1.11, 5.48], whereas its normalized contrast was 1.05 pp [-1.19, 2.96]. The residual fixed response was smaller than the native response but remained positive. Because eligibility stability is conditioned after replay, these estimates have a different, potentially selected estimand.

First-divergence anatomy showed the corresponding change in pathway composition. Eligibility boundaries accounted for 67.09% [63.02%, 71.07%] of fixed-pixel identity exchanges and 51.01% [46.06%, 55.94%] under normalized replay. Within-set geometric rank reversals increased from 30.13% [26.14%, 34.36%] to 45.37% [40.62%, 50.26%]. These corrected v2 proportions supersede the earlier anatomy values.

Table 2

Restricted mean stability radius through $\tau = 0.125$. Intervals use 5,000 stratified image-cluster bootstrap replicates.

Quantity	Point	95% CI
O2O RMSR	0.11992	[0.11940, 0.12038]
O2M assigned-positive RMSR	0.10055	[0.09895, 0.10210]
Tiny O2O–O2M RMSR	0.01724	[0.01565, 0.01876]
Small O2O–O2M RMSR	0.03282	[0.02904, 0.03614]

The aggregate O2O branch response was already present before final conflict resolution. The loss-active-minus-pre-conflict scale-gap difference was -0.077 pp [-0.367, 0.313] for fixed replay and -0.395 pp [-0.986, 0.034] for normalized replay. Object-level fragility disagreement was 0.578% under both contracts. All 134 fixed directional disagreements and all 96 normalized disagreements coincided with the instrumented conflict stage. Conflict therefore localised the small set of pre/post-resolution disagreements without explaining the aggregate branch-scale pattern or establishing a causal resolver effect.

These boundary-localisation results are shown in Figure 3. The eligibility-stable analysis and eligibility-lock counterfactual are deliberately presented as distinct estimands.

4.4. O2O and O2M occupy different local assignment-boundary geometries

Continuous-radius analysis included 29,724 directional trajectories from 7,431 focal GTs. Through $\tau = 0.125$, the O2O loss-active identity had RMSR 0.11992, compared with 0.10055 for the O2M assigned-positive Top-1 state (Table 2). The O2O-minus-O2M RMSR contrast was 0.01724 for 8–16 px objects and 0.03282 for 16–32 px objects. By τ , an O2O boundary had been observed for 9.03% [8.26%, 9.90%] of tiny and 6.39% [4.61%, 8.49%] of small trajectories. The corresponding O2M rates were 32.66% [31.19%, 34.14%] and 48.15% [44.79%, 51.31%]. O2O and O2M therefore differed in both overall boundary proximity and its scale structure.

O2M rank-set decomposition clarified the reverse normalized scale response. Under fixed replay, tiny-minus-small Jaccard and base-retention contrasts were -4.19 and -2.26 pp. Under normalized replay, they reversed to +5.91 and +4.93 pp, indicating lower assigned-positive-set persistence for 16–32 px objects. The normalized assigned-set Top-1 flip contrast was -12.80 pp, while the rank-only share

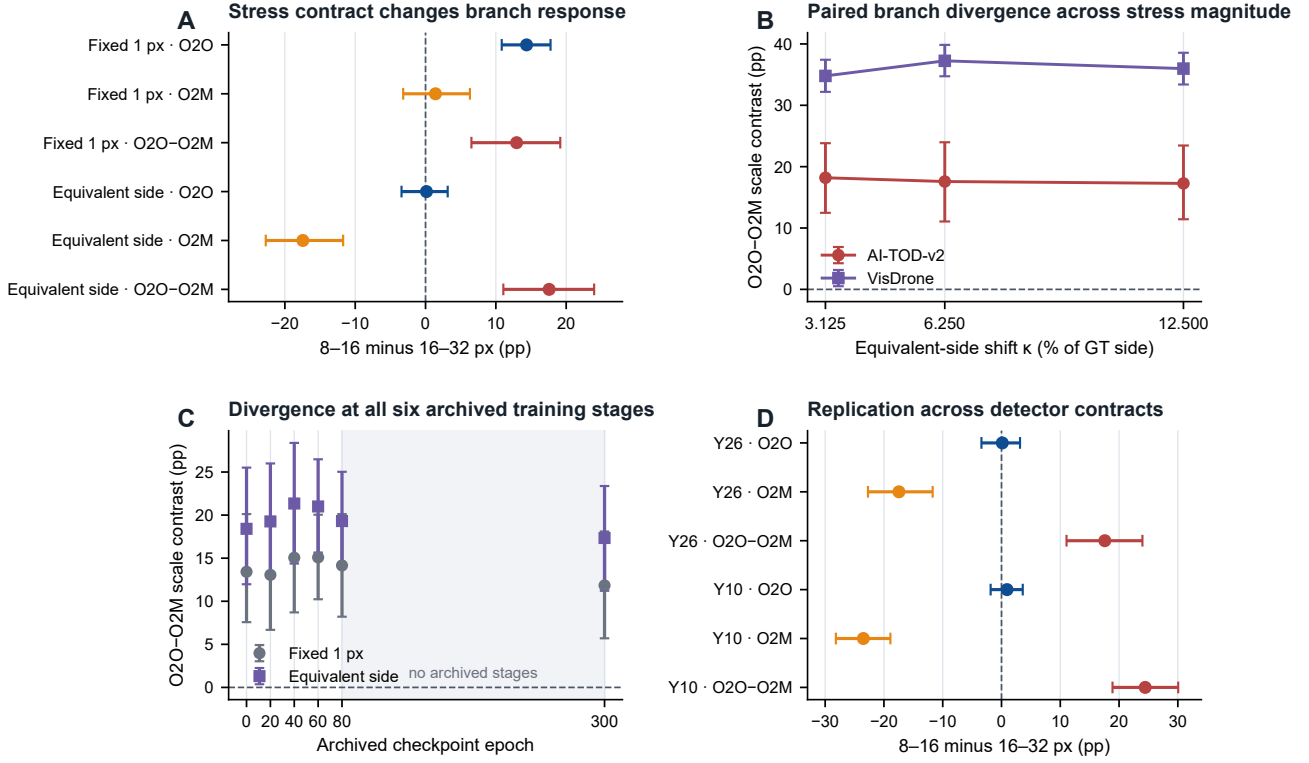


Figure 2: Stress-contract and branch-response structure. (A) AI-TOD-v2 scale contrasts under fixed one-pixel and equivalent-side replay. (B) Paired O2O-minus-O2M contrasts across the three prespecified equivalent-side magnitudes on AI-TOD-v2 and VisDrone. (C) Paired contrasts at all six physically archived training stages; points are not connected, and the shaded 80–300 interval contains no archived intermediate checkpoints. (D) Equivalent-side replication under compatible YOLO26s and YOLOv10-S detector contracts. Points are inverse-inclusion-weighted estimates; whiskers are 95% intervals from 5,000 stratified image-cluster bootstrap replicates. Positive scale contrasts denote higher 8–16 px than 16–32 px fragility.

Table 3

Tiny-minus-small O2M rank-set contrasts in percentage points. Full intervals and empty-set transitions are reported in the Supplementary Material.

Contract	Δ Jaccard	Δ retention	Δ Top-1 flip	Δ rank-only share
Fixed 1 px	-4.19	-2.26	-2.38	-10.10
Equivalent-side	5.91	4.93	-12.80	22.50

among flips was +22.50 pp. Smaller objects therefore had fewer assigned-set Top-1 flips but a larger rank-only fraction among those flips; larger objects combined greater Top-1 turnover with lower set persistence. Rank state and positive-set state cannot be treated as one O2M identity variable.

The strict same-stride analysis retained 3,835 paired GTs. Its O2O scale contrast was +10.23 pp [7.30, 13.32], its O2M contrast was -0.12 pp [-6.83, 5.78], and the paired difference was +10.35 pp [3.54, 18.17]. Cross-stride switching alone therefore did not account for the branch contrast, although post-replay conditioning can induce selection and does not exclude a feature-pyramid-level contribution.

Figure 4 complements the endpoint contrasts with continuous identity-survival, cause-specific incidence, RMSR, and O2M Rank-Set geometry.

4.5. Native margin measures local rank separation, with a narrower utility boundary

Only 1,592 of 29,724 trajectories had an observed fixed-pair q -order crossing; 7,437 were competing-censored, 16,551 right-censored, and 4,144 undefined. Within the observed-crossing subset, margin correlated with crossing radius at $\rho = 0.628$ [0.584, 0.670]. Its correlation with observed eligibility-boundary radius was $\rho = 0.153$ [0.114, 0.194], a difference of 0.475 [0.405, 0.540]. Margin therefore had construct validity as a local rank-separation coordinate, but the result is conditional on observing a crossing and is not a censoring-adjusted population association.

Replacing a linear margin term with a training-fold-fitted restricted cubic spline did not improve held-out metrics consistently. For fragility, spline-minus-linear ROC-AUC and PR-AUC differences were -0.00756 [-0.01411, -0.00138]

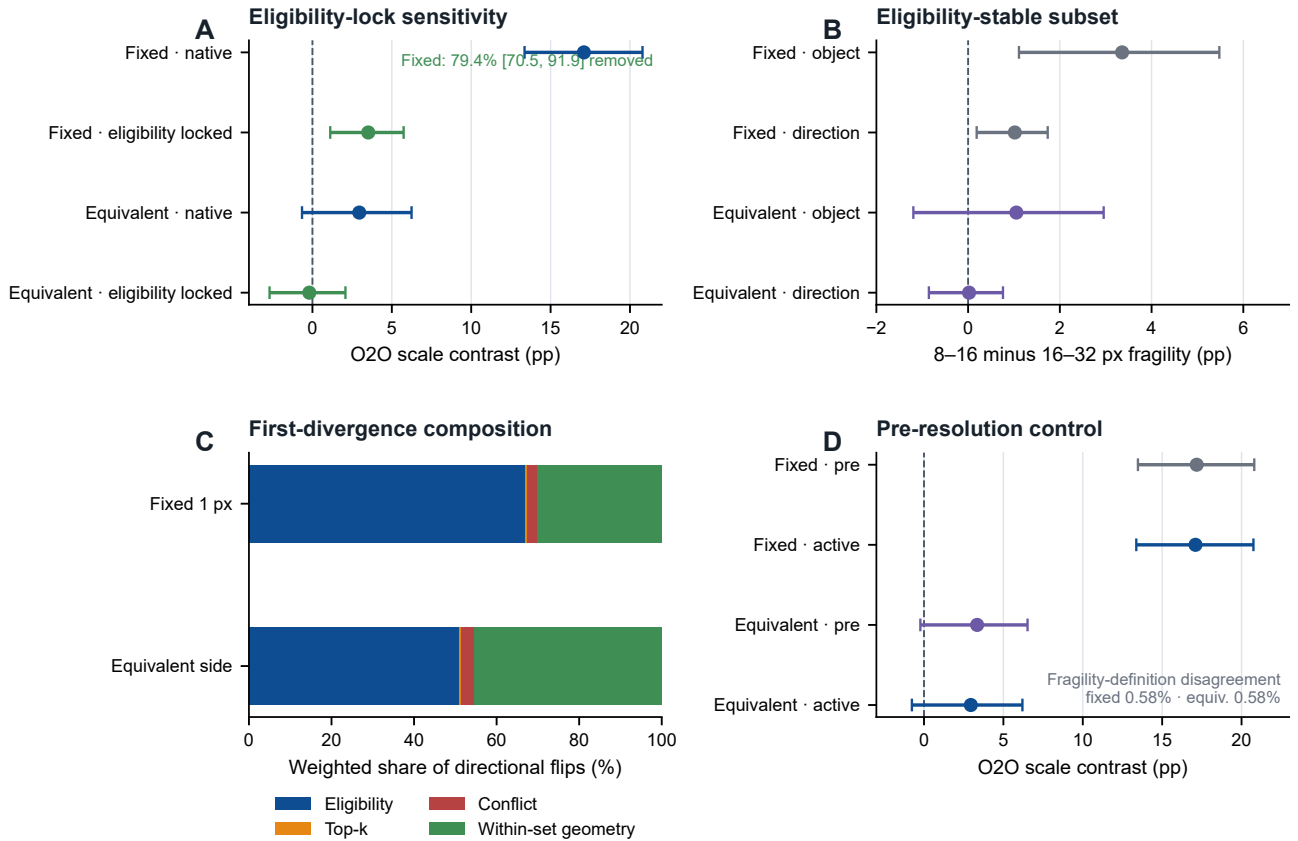


Figure 3: Eligibility and native-pathway localisation. (A) Native versus base-eligibility-locked O2O scale contrasts. Eligibility locking removed 79.4% [70.5%, 91.9%] of the fixed-pixel gap. (B) Object- and direction-level contrasts conditional on eligibility stability. (C) Mutually exclusive first-divergence shares among 2,194 fixed and 1,241 equivalent-side directional O2O identity exchanges; geometry-clipping events are overlapping diagnostics and therefore are not stacked. Equivalent-side minus fixed shares were -16.1 pp [-18.9 , -13.3] for eligibility and $+15.2$ pp [$+12.7$, $+17.8$] for within-set geometry. (D) Pre-conflict Top-7 winner versus final loss-active scale contrasts. Whiskers are 95% intervals from 5,000 stratified image-cluster bootstrap replicates. First divergence localises the earliest instrumented state difference and is not a causal attribution.

and -0.00415 [-0.00813 , -0.00109]. Coverage-based-miss PR-AUC was also lower by -0.00541 [-0.01298 , -0.00027], while the log-loss intervals spanned zero for both outcomes. This negative sensitivity result does not establish equivalence or identify a best predictor; it limits claims that a more flexible margin shape supplies additional diagnostic utility.

Figure 5 separates margin's construct validity from its more limited incremental diagnostic utility.

4.6. Locality, efficiency, and the confirmation boundary

Focal replay changed a non-focal O2O identity in 52 of 3549250 pair-events. The overall weighted pair rate was 0.001465% [0.000824% , 0.002383%]. All observed changes occurred among the 5,679 pairs sharing at least one native pre-conflict candidate claim; their rate was 0.9159% [0.5787% , 1.3248%]. No event was observed among 3,543,571 nonsharing pairs in this sample. The response was therefore rare and structurally concentrated, but zero observed events do not establish absence, equivalence, or exclusivity.

At a 100-image audit budget, all 1,000 subsamples recovered the sign and O2O-versus-O2M branch ordering of the full normalized paired contrast. The median absolute error was 3.43 pp and the 95th-percentile error was 8.41 pp. Reduced budgets can screen a dominant response signature, but they do not provide precise effect estimates or a universal sample-size rule.

The preregistered disjoint-sample confirmation produced no normative verdict. The single permitted analyzer invocation wrote the irreversible outcome-access receipt and then terminated before any H1–H5 estimate or decision because its support assertions preceded the specified zero-cluster and common-support operations. It was not rerun. A post-access corrective analysis retained all 300 image clusters and used the 7243-key fixed/normalized common-valid intersection, with 3 fixed-only keys disclosed. All five exploratory contrast directions agreed with the preregistration, but every formal status remained invalidated and not retested. The full chronology and exploratory intervals are reported in the Supplementary Material.

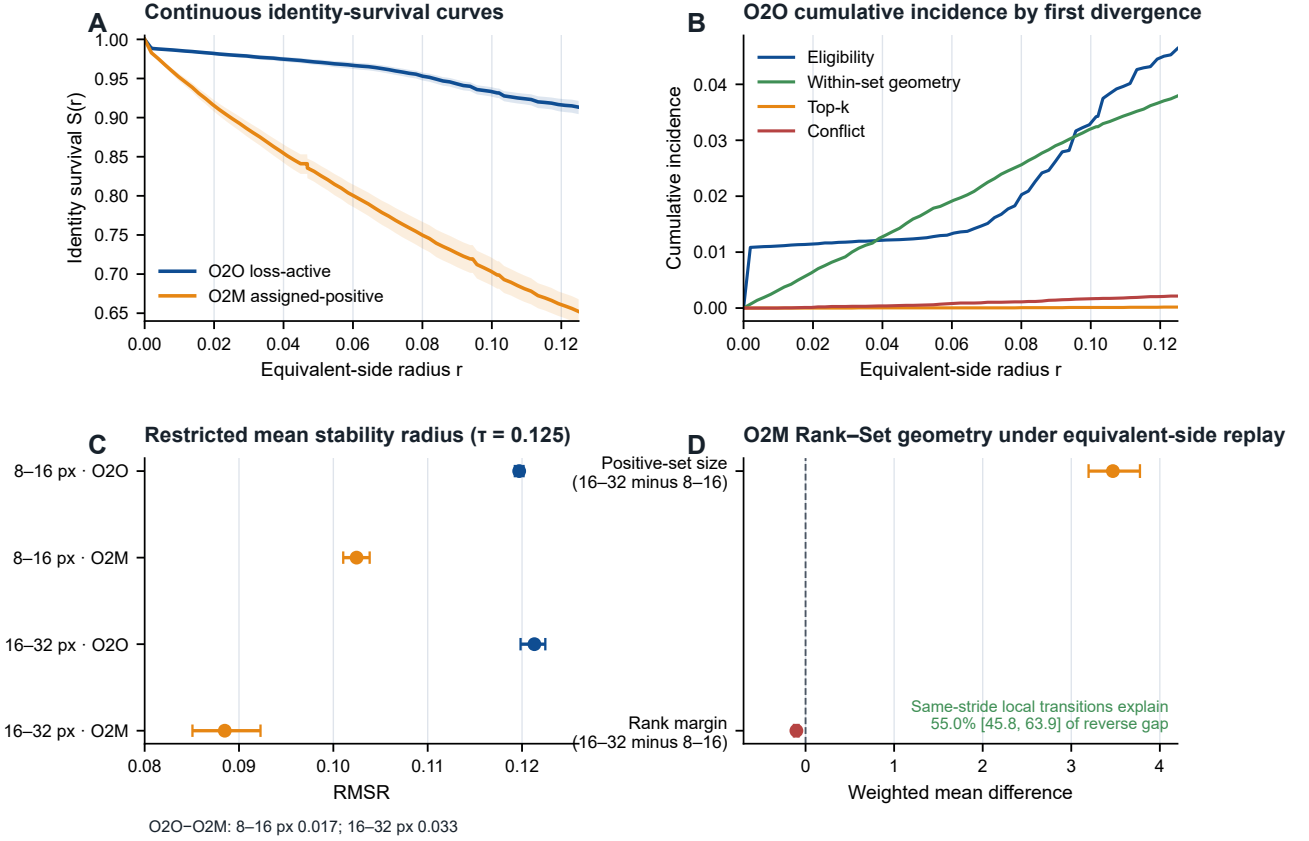


Figure 4: Continuous assignment-boundary geometry and O2M Rank-Set structure. (A) Weighted Kaplan-Meier identity-survival curves for O2O loss-active and O2M assigned-positive Top-1 states. (B) Aalen-Johansen cumulative incidence of the instrumented O2O first-divergence causes. (C) Restricted mean stability radius through $\tau = 0.125$ for 8-16 px and 16-32 px objects. (D) AI-TOD-v2 normalized O2M positive-set-size and rank-margin differences (16-32 minus 8-16 px); the text reports the prespecified same-stride-local share of the reverse directional gap. Shading and whiskers are 95% intervals from 5,000 stratified image-cluster bootstrap replicates.

5. Discussion

5.1. Stress response is a property of the measurement contract

The fixed-pixel O2O scale gradient and its attenuation under equivalent-side replay should not be treated as competing estimates of one intrinsic quantity. Fixed replay applies a larger fractional displacement to a smaller object; normalized replay controls that fraction but changes the question. Their difference is informative because it identifies contract dependence. The stable positive paired branch contrast across the tested κ grid, datasets, archived checkpoints, and a compatible YOLOv10-S contract is therefore more general than any one absolute O2O scale contrast, but it still describes the audited contracts rather than an architecture-invariant law.

Eligibility locking explains why the fixed and normalized contracts differ. Most of the fixed O2O gap disappeared when the base eligibility state was held constant, and the first-divergence share assigned to eligibility fell under normalized replay. At the same time, within-set rank reversals became more prominent. The evidence supports a layered

interpretation: fixed-pixel stress often encounters a hard candidate-admission boundary, whereas normalized stress leaves a larger fraction of surviving exchanges to geometric competition among already comparable candidates. This is pathway localisation, not proof that eligibility alone causes the complete scale response.

5.2. O2O identity, O2M rank, and O2M set persistence are different states

Continuous boundary analysis changes the interpretation of the five-state results. O2O loss-active identity had a larger restricted mean stability radius than the audited O2M assigned-set Top-1 state, while O2M boundary incidence increased from tiny to small objects. The branch contrast is therefore not only a difference between two percentages at one chosen displacement; it reflects different local boundary geometries over a range of equivalent-side radii.

Rank-set decomposition shows why one O2M identity label is insufficient. Under normalized stress, 16-32 px objects had lower assigned-positive-set persistence and more Top-1 turnover. Conversely, the fewer tiny-object Top-1 flips had a larger rank-only share. Set turnover and rank exchange

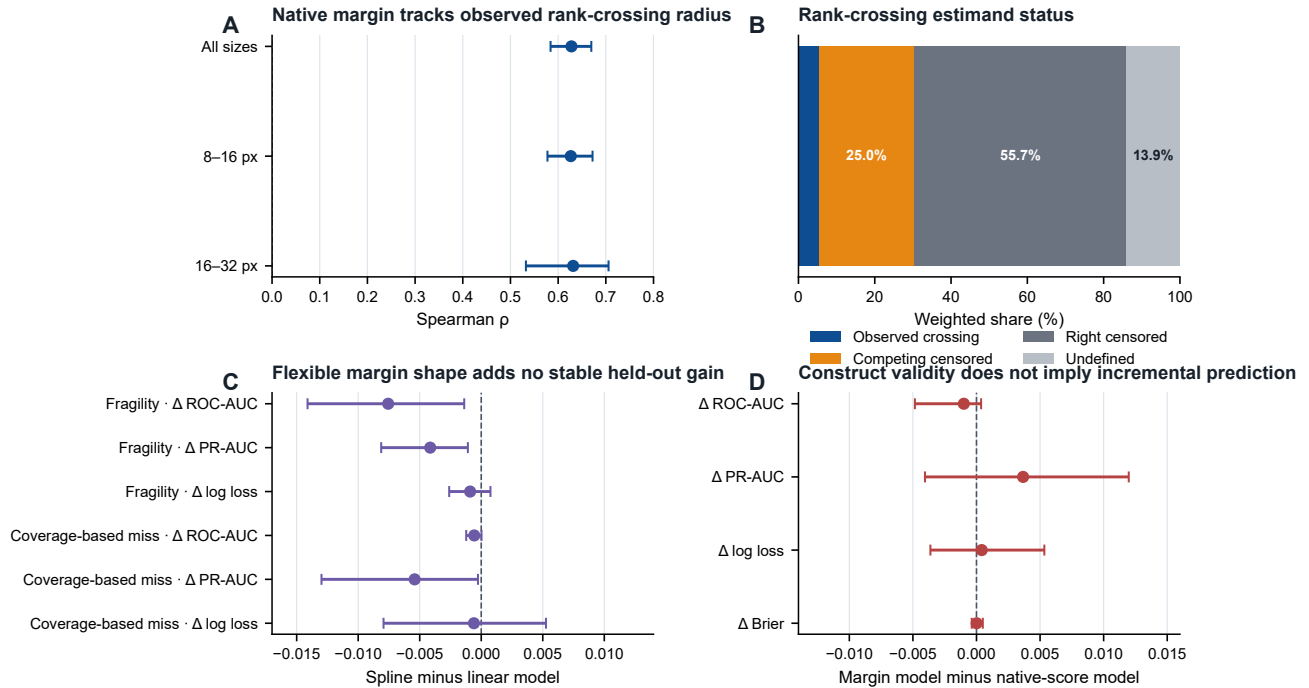


Figure 5: Margin construct validity and utility boundary. (A) Spearman association between native active-relative margin and observed fixed-pair rank-crossing radius, conditional on an observed crossing. (B) Weighted status composition of the rank-crossing estimand; competing censoring, right censoring, and undefined states prevent a population-wide uncensored interpretation. (C) Held-out metric differences between restricted-cubic-spline and linear margin specifications. (D) Out-of-fold increment from adding margin after area, O2M coverage, native active score, and active ClO u for the coverage-based-miss outcome. Whiskers are 95% stratified image-cluster bootstrap intervals. Association supports margin as a rank-separation coordinate; it does not establish stable incremental prediction or treatment efficacy.

can therefore move together or separately. This distinction also sharpens the limitation of same-model O2O–O2M comparisons: the branches differ in supervision semantics, but those differences can be decomposed into explicit native states instead of being left as an unspecified confounder.

The O2O pre-resolution control places conflict resolution in the same hierarchy. Pre-conflict and final loss-active scale gaps were nearly identical, and object-level disagreement was rare. Every observed disagreement coincided with the conflict stage, which localises where the final identity changed. However, coincidence under a deterministic path does not estimate a causal resolver effect, and it does not imply that conflict is unimportant in other checkpoints or detector contracts.

5.3. Margin has construct validity for rank separation, not for all instability

The continuous analysis identifies a narrower meaning for native margin. Among trajectories with an observed crossing for the fixed active/runner pair, larger margin was associated with a more distant rank boundary. Its association with eligibility-boundary radius was much weaker. Margin therefore measures one component of assignment geometry: separation between a particular comparable pair. It does not measure whether either candidate will remain eligible,

whether a conflict will change ownership, or whether a valid runner-up exists.

This construct interpretation does not automatically confer downstream utility. A more flexible spline representation did not improve held-out ROC-AUC or PR-AUC over the linear-margin model and did not produce a consistent log-loss advantage. The proper evidence chain is thus construct localisation without a claim of causal, predictive, or treatment validity. Undefined competition remains especially important: when no legal relative comparison exists, margin has no value rather than an extreme value.

5.4. Local coupling and practical audit use

Spillover was rare across all focal–non-focal pairs but concentrated among pairs sharing a native pre-conflict candidate claim. This supports a local structural coupling interpretation of the assignment path. The absence of observed nonsharing events in this sample is not evidence of equivalence or global independence; a different sample could contain rare nonlocal coupling.

The sample-efficiency analysis suggests a bounded engineering use. One hundred images consistently recovered the dominant sign and branch order, but the upper-tail estimation error remained large. Reduced audits may therefore screen checkpoints for a strong stress signature, whereas effect estimation and scientific comparison require the full design.

The audit is not a certification threshold, an acceptance test, or a runtime uncertainty score.

5.5. Relation to assignment and diagnostic work

Tiny-object assignment methods change similarity, positive construction, density, or supervision schedules [1, 2, 3, 4, 13]. O2O/O2M detector designs similarly modify matching objectives or add auxiliary supervision [6, 7, 9, 8]. The present framework does not propose another assignment rule. It measures the resolved native state under controlled replay and separates identity, rank, set, and boundary-cause responses.

Output-level diagnostics such as TIDE decompose detection errors after inference [32]. The present audit instead targets the training assignment state, while keeping inference and frozen network outputs unchanged. These perspectives are complementary: one locates output error categories, and the other locates instability inside the native supervision path.

5.6. Prospective confirmation boundary

The attempted lockbox does not provide confirmatory replication. Instrument freeze, preregistration, disjoint selection, and four sealed extractions completed, but the only permitted analyzer invocation failed after the irreversible receipt and before any preregistered estimate or verdict. The prospective procedure was therefore invalidated rather than passed, failed statistically, or found nonsignificant. Not rerunning it preserves the one-time rule.

The corrected post-access directions were concordant with the preregistration, but the support rule was specified after outcome access. Those estimates remain useful as transparent exploratory sensitivity and as evidence for repairing future instruments; they cannot recover prospective status. The discovery findings remain content-addressed observations, but their confirmatory strength is lower than it would have been with a valid independent verdict.

6. Limitations

The audit is diagnostic and read-only. It measures the response of frozen native assignment states to controlled geometric replay; it does not identify a causal training mechanism, estimate annotation noise, or show that changing the measured state would improve detection. The O2O–O2M comparison also remains observational because the branches use different learned scores, positive cardinalities, and supervision semantics.

Generalisability is bounded. AI-TOD-v2 provided the principal discovery sample, and VisDrone, three normalized magnitudes, six archived YOLO26s stages, and one compatible YOLOv10-S contract supplied sensitivity or replication evidence. The six checkpoint files do not cover epochs 100–280, so the trajectory cannot establish continuous persistence throughout training. The first-divergence proportions were estimated on one AI-TOD-v2 seed-0 checkpoint and should not be assumed invariant across datasets, seeds, or detector contracts.

The eligibility-stable and same-stride analyses condition on observed replay states. They address specific alternative explanations but may induce selection. The pre-resolution disagreement analysis localises changes to the conflict stage without estimating its causal contribution. Spillover was rare and no event was observed among nonsharing pairs in this sample, but zero observed events do not establish independence, exclusivity, or equivalence.

Continuous-boundary results are limited to equivalent-side cardinal trajectories through $r_{\max} = 0.25$ and RMSR truncation $\tau = 0.125$. A first boundary outside that range is right-censored. Margin- ρ_R association is conditional on the 1,592 observed fixed-pair crossings and is not a censoring-adjusted population correlation. Boundary search changes GT geometry while keeping features frozen; it does not model image formation under physical camera motion.

The coverage-based-miss diagnostic is not the standard one-to-one evaluator false negative because predictions are not consumed. The spline sensitivity compares two specified margin shapes and does not prove equivalence or exclude other predictors. Reduced-budget results estimate screening error under the frozen stratified sample and do not define a universal sample-size rule.

Finally, the preregistered lockbox yielded no normative verdict. The single analyzer invocation was irreversibly opened and then invalidated by a support-contract assertion before H1–H5 estimates or decisions were written. Prospective rerun was prohibited. The post-access correction retained all 300 clusters and disclosed the 7,243-key intersection, but its directional agreement is exploratory only and cannot be described as confirmation.

7. Conclusion

We developed a qualified read-only framework for measuring the local geometry of native dense-detector assignment states. The results distinguish three levels of stability: hard eligibility boundaries, within-set rank boundaries, and multi-positive-set persistence. Fixed-pixel O2O scale responses were largely coupled to eligibility boundaries, whereas normalized replay reduced that component and increased the share of within-set geometric exchanges. Continuous analysis further showed that O2O loss-active identity and O2M assigned-positive rank occupy different stability-radius geometries, while Rank–Set decomposition separated O2M rank turnover from positive-set turnover.

Native margin reflected one part of this structure, the distance to an observed fixed-pair rank crossing, but not eligibility, missing competition, or every form of assignment instability. The framework is a diagnostic-testing interface for native assignment-state geometry, not a replacement assigner, calibrated failure probability, runtime uncertainty estimator, or validated intervention. The attempted prospective audit produced no confirmatory verdict; its post-access correction remains exploratory. Future work should replicate these decompositions under additional frozen detector contracts.

CRedit authorship contribution statement

Yede Liu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing—original draft. **Fang Wang:** Validation, Investigation, Writing—review and editing. **Jun Li:** Supervision, Project administration, Writing—review and editing.

Declarations

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Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability. The [AI-TOD-v2](#) and [VisDrone](#) repositories provide the source datasets; raw images and annotations are not redistributed under their terms. Audit code, outcome-blind sampling manifests, derived per-object tables, bootstrap summaries, figure source data, and checkpoint SHA-256 identifiers are publicly available in the [project repository](#). Model binaries, raw prediction dumps, and third-party images are not redistributed.

Declaration of generative AI and AI-assisted technologies in the writing process. During preparation, the authors used OpenAI Codex (OpenAI) for code assistance, statistical-workflow implementation, language editing, plotting code, and figure-layout refinement. All analyses, result verification, scientific judgments, and final interpretations were reviewed and approved by the authors, who take full responsibility for the article.

References

- [1] J. Wang, C. Xu, W. Yang, and L. Yu, "A normalized Gaussian Wasserstein distance for tiny object detection," *arXiv preprint arXiv:2110.13389*, 2021, doi:10.48550/arXiv.2110.13389.
- [2] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *European Conference on Computer Vision (ECCV)*, pp. 526–543, 2022, doi:10.1007/978-3-031-20077-9_31.
- [3] S. Shi, Q. Fang, X. Xu, and T. Zhao, "Similarity distance-based label assignment for tiny object detection," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 13711–13718, 2024, doi:10.1109/IROS58592.2024.10801448.
- [4] C. Xu, J. Ding, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Dynamic coarse-to-fine learning for oriented tiny object detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7318–7328, 2023.
- [5] H. Sun, Y. Li, L. Yang, X. Cao, and B. Zhang, "Uncertainty-aware gradient stabilization for small object detection," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 8407–8417, 2025.
- [6] J. Wang, L. Song, Z. Li, H. Sun, J. Sun, and N. Zheng, "End-to-end object detection with fully convolutional network," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 15849–15858, 2021.
- [7] P. Sun, Y. Jiang, E. Xie, W. Shao, Z. Yuan, C. Wang, and P. Luo, "What makes for end-to-end object detection?" in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139, pp. 9934–9944, 2021.
- [8] Y. Chen, Q. Chen, Q. Hu, and J. Cheng, "Dual assignment of labels for end-to-end fully convolutional object detection," *Pattern Recognition*, vol. 169, art. 111877, 2026, doi:10.1016/j.patcog.2025.111877.
- [9] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, doi:10.52202/079017-3429.
- [10] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLOv26: Unified real-time end-to-end vision models," *arXiv preprint arXiv:2606.03748*, 2026, doi:10.48550/arXiv.2606.03748.
- [11] P. Liu, S. Shang, C. Song, F. Li, Y. Guo, Y. Chang, H. Feng, G.-S. Xia, B. Wu, J. Tan, and W. Wang, "TCD: Towards consistent and unified single stage object detection," *Neurocomputing*, vol. 671, art. 132720, 2026, doi:10.1016/j.neucom.2026.132720.
- [12] G. Cao, X. Lan, W. Huang, J. Zhang, D. Jiang, and Y. Wang, "DS-Det: Single-query paradigm and attention disentangled learning for flexible object detection," in *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 286–295, 2025, doi:10.1145/3746027.3755045.
- [13] C. Liu, Y. Chen, N. Yin, G. Zhao, and X. Qian, "Density-aware adaptive label assignment for end-to-end dense detection in drone images," *Expert Systems with Applications*, vol. 331, art. 133189, 2026, doi:10.1016/j.eswa.2026.133189.
- [14] Y. Ma, J. Li, L. Yang, Y. Su, Y. Li, and W. Yang, "Shadow-DETR: Alleviating matching conflicts through shadow queries," *Neural Networks*, vol. 197, art. 108524, 2026, doi:10.1016/j.neunet.2025.108524.
- [15] S. Lin, L. Zhong, S. Chen, and D.-H. Wang, "Tiny object detection via normalized Gaussian label assignment and multi-scale hybrid attention," *Remote Sensing*, vol. 18, no. 3, art. 396, 2026.
- [16] J. Zhang, Q. Fang, S. Shi, Y. Zeng, and X. Xu, "VALA: Virtual anchor-guided label assignment for tiny object detection," *Neurocomputing*, vol. 696, art. 134055, 2026, doi:10.1016/j.neucom.2026.134055.
- [17] A. Tian, A. Wang, and C. Fu, "NBD-DLA: Normalized Bhattacharyya distance-based dynamic label assignment for remote sensing tiny object detection," *Neurocomputing*, art. 134665, 2026, doi:10.1016/j.neucom.2026.134665.
- [18] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9759–9768, 2020, doi:10.1109/CVPR42600.2020.00978.
- [19] K. Kim and H. S. Lee, "Probabilistic anchor assignment with IoU prediction for object detection," in *European Conference on Computer Vision (ECCV)*, pp. 355–371, 2020, doi:10.1007/978-3-030-58595-2_22.
- [20] Z. Ge, S. Liu, Z. Li, O. Yoshie, and J. Sun, "OTA: Optimal transport assignment for object detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 303–312, 2021.
- [21] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3510–3519, 2021, doi:10.1109/ICCV48922.2021.00349.
- [22] X. Zhang, F. Wan, C. Liu, R. Ji, and Q. Ye, "FreeAnchor: Learning to match anchors for visual object detection," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [23] B. Zhu, J. Wang, Z. Jiang, F. Zong, S. Liu, Z. Li, and J. Sun, "AutoAssign: Differentiable label assignment for dense object detection," *arXiv preprint arXiv:2007.03496*, 2020, doi:10.48550/arXiv.2007.03496.
- [24] C. Zhang, Z. Chen, B. Xiong, K. Ji, and G. Kuang, "EOOD: End-to-end oriented object detection," *Neurocomputing*, vol. 621, art. 129251, 2025, doi:10.1016/j.neucom.2024.129251.
- [25] D. Jia, Y. Yuan, H. He, X. Wu, H. Yu, W. Lin, L. Sun, C. Zhang, and H. Hu, "DETRs with hybrid matching," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19702–19712, 2023.
- [26] Q. Chen, X. Chen, J. Wang, S. Zhang, K. Yao, H. Feng, J. Han, E. Ding, G. Zeng, and J. Wang, "Group DETR: Fast DETR training with group-wise one-to-many assignment," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6633–6642, 2023.
- [27] Z. Zong, G. Song, and Y. Liu, "DETRs with collaborative hybrid assignments training," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6748–6758, 2023.
- [28] Z. Kang, J. Zhuang, K. Mo, Q. Chen, R. Liu, and Y. Zhang, "PaQ-DETR: Learning pattern and quality-aware dynamic queries for object detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25504–25513, 2026.
- [29] S. Liu, T. Ren, J. Chen, Z. Zeng, H. Zhang, F. Li, H. Li, J. Huang, H. Su, J. Zhu, and L. Zhang, "Detection transformer with stable matching," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6491–6500, 2023.
- [30] K. N. A. Shastry, K. R. S. Teja, A. Nigam, and C. Arora, "Favoring one among equals—not a good idea: Many-to-one matching for robust transformer based pedestrian detection," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 748–757, 2024, doi:10.1109/WACV57701.2024.00081.
- [31] S. Li, M. Li, R. Li, C. He, and L. Zhang, "One-to-few label assignment for end-to-end dense detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7350–7359, 2023.
- [32] D. Bolya, S. Foley, J. Hays, and J. Hoffman, "TIDE: A general toolbox for identifying object detection errors," in *European Conference on Computer Vision (ECCV)*, pp. 558–573, 2020.
- [33] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 190, pp. 79–93, 2022, doi:10.1016/j.isprsjprs.2022.06.002.
- [34] P. Zhu et al., "Vision meets drones: A challenge," *arXiv preprint arXiv:1804.07437*, 2018, doi:10.48550/arXiv.1804.07437.
- [35] Q. Hu, Y. Guo, X. Xie, M. Cordy, L. Ma, M. Papadakis, and Y. Le Traon, "Test optimization in DNN testing: A survey," *ACM Transactions on Software Engineering and Methodology*, vol. 33, no. 4, art. 111, pp. 1–42, 2024, doi:10.1145/3643678.
- [36] Z. Chen, W. Luo, S. Dai, Q. Yang, B. Guo, and X. Lin, "Test case selection via discrepant features amplification for deep neural networks," *Neurocomputing*, vol. 672, art. 132793, 2026, doi:10.1016/j.neucom.2026.132793.
- [37] E. Kaufman, A. Levy, Y. Adler, A. Levi, and M. Sinai, "Near misses analysis (NMA): A new explainable AI approach for model understanding, comparison, debugging and adversarial attack detection," *Neurocomputing*, vol. 694, art. 133875, 2026, doi:10.1016/j.neucom.2026.133875.
- [38] V. Moustakidis, N. Passalis, and A. Tefas, "Detecting and mitigating training anomalies in deep neural networks," *Neurocomputing*, vol. 702, art. 134666, 2026, doi:10.1016/j.neucom.2026.134666.